

Object-Conditioned Strategy-Level Parameter Configuration for Haptic Teleoperation of Heterogeneous Objects

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Abstract

Fixed teleoperation parameters may not provide suitable interaction behavior across objects with different fragility, geometry, and grasping requirements. Existing object-conditioned approaches mainly adjust slave-side impedance, whereas joint configuration of slave-side impedance, master-side haptic feedback, and gripper execution parameters has received limited attention. To address this issue, this paper proposes an object-conditioned strategy-level parameter-configuration framework for haptic teleoperation of heterogeneous objects. Each detected object class is mapped to one of three manipulation strategies, which jointly specify the parameter settings of the three channels. Low-rate visual perception is integrated asynchronously with a nominal 200-Hz supervisory loop. The first mapped detection satisfying the confidence threshold triggers a one-shot parameter update, and the selected strategy remains locked for the remainder of the trial. The framework was evaluated in a human-in-the-loop experiment involving three operators, six objects, five control modes, and 135 grasp-and-transfer trials arranged in 27 matched blocks. Compared with impedance-only reconfiguration, the complete joint configuration reduced mean task completion time by 1.79 s (8.5%; matched-block bootstrap 95% CI, 1.10–2.51 s). The improvement was directionally consistent across operators and objects, while secondary outcomes were analyzed descriptively without attributing gains to any single parameter channel. Overall, the proposed framework provides an interpretable one-shot configuration approach for haptic teleoperation of heterogeneous objects.

Keywords: haptic teleoperation, heterogeneous objects, impedance control, object-conditioned configuration, haptic feedback, human-in-the-loop evaluation

1. Introduction

Teleoperation combines human decision-making with robotic execution, enabling remote robots to perform complex manipulation tasks that remain challenging for fully autonomous systems, including hazardous operations, flexible manufacturing, and manipulation in semi-structured or unstructured environments. Haptic feedback conveys remote interaction information to the operator, supporting the perception of contact conditions, grasp stability, and potential object slip. Effective teleoperation therefore requires the coordinated integration of bilateral control, environmental perception, robotic execution, and human-machine interaction [1, 2].

Teleoperation becomes particularly challenging when objects with heterogeneous interaction requirements must be manipulated within the same system. These objects may differ substantially in geometry, deformability, contact sensitivity, and grasp-stability requirements, making a single fixed parameter configuration difficult to reconcile with both compliant contact and stable manipulation. For fragile or deformable objects, lower robot stiffness and requested grasp force may reduce contact impact, but can also compromise positioning accuracy and grasp stability. Conversely, higher stiffness or greater requested grasp

force may improve pose retention and grasp stability for rigid or slip-prone objects, but may also increase contact impact or deformation risk. Object categories can therefore serve as compact semantic indicators of task-related interaction requirements and support object-dependent parameter configuration.

Impedance control provides a fundamental approach for regulating physical interaction by shaping the dynamic relationship between robot motion and interaction force [3]. Variable-impedance control has been extensively investigated to improve robot adaptability by adjusting stiffness and damping according to task state, contact information, human behavior, or learned policies [4, 5]. In teleoperation, variable-impedance and bilateral-control approaches have been developed to regulate remote-robot behavior according to operator motion, estimated human impedance, contact state, task phase, and environmental interaction [6–13]. These studies have provided important foundations for improving remote compliance, interaction stability, and teleoperation transparency.

In parallel, advances in visual and multimodal perception have enabled teleoperation systems to incorporate object- and task-related information into control and interaction strategies. Previous studies have employed visual information, force feedback, speech, gaze, and other modalities to select task-dependent impedance parameters, generate interaction strategies, and assist operator decision-making [14–20]. Tactile and kinesthetic feedback, together with grip-safety information, have also been

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investigated to support operator awareness and grasp execution [21–24]. Collectively, these studies indicate that object- and task-level information can support parameter selection and strategy adaptation in teleoperation.

Despite these advances, many prior approaches have focused on individual aspects of teleoperation, particularly slave-side impedance adaptation or control-authority allocation [6–13, 16, 19, 20]. Teleoperation performance, however, depends jointly on remote-robot interaction, the haptic information presented to the operator, and gripper execution. Slave-side impedance regulates contact compliance, master-side haptic feedback shapes the operator’s perception of remote interaction, and gripper parameters govern the execution of grasp commands. Although these channels affect different aspects of task execution, they play complementary roles in task efficiency, contact behavior, and grasp stability. This observation motivates the use of object-level semantic information to establish an interpretable manipulation strategy and coordinate the initial configuration of multiple task-related channels while retaining operator control.

Direct integration of visual-semantic inference into the high-rate teleoperation loop introduces an additional system-integration challenge. Visual inference generally operates at a lower update rate and may be subject to computational latency and detection fluctuations, whereas the teleoperation loop must remain continuously responsive to operator input. Using each perception result to trigger an immediate parameter update could delay supervisory processing, cause repeated switching, and produce inconsistent interaction behavior. A practical architecture should therefore translate visual-semantic information into a stable task-level strategy without placing visual inference in the high-rate supervisory path.

To address these configuration and integration issues, this paper proposes an **object-conditioned strategy-level configuration framework** for haptic teleoperation of heterogeneous objects in semi-structured manipulation scenarios. Each detected object class is interpreted in terms of its task-relevant interaction requirements and mapped to one of three manipulation strategies: *fragility-priority*, *balanced*, and *stability-priority*. The selected strategy coordinates three task-related parameter channels: slave-side impedance, master-side haptic feedback, and gripper execution. The object-to-strategy mapping forms an interpretable intermediate layer between visual recognition and multi-channel parameter configuration. The corresponding operating points were selected through engineering-based qualitative screening for the tested objects and platform.

Following the first threshold-qualified result read by the supervisory loop, the proposed framework performs a one-shot strategy-level parameter update and retains the selected strategy for the remainder of the trial. This one-shot, strategy-locked design decouples low-rate visual-semantic inference from the nominal 200-Hz supervisory loop. It prevents repeated switching caused by detection fluctuations and maintains a consistent parameter configuration while retaining operator control.

The framework was evaluated on an Omega.7–Franka Panda platform using three operators, six objects, five modes, and 135 trials arranged in 27 matched blocks. Task completion time was the primary endpoint, with the primary comparison between

complete multi-channel configuration (mode C) and impedance-only reconfiguration (mode E). Modes A, B, and D provided fixed-parameter, timed manual-selection, and semantic-display reference conditions, respectively.

The main contributions of this work are summarized as follows:

1. **Interpretable object-to-strategy mapping.** An interpretable intermediate strategy layer translates object-level semantic information into coordinated operating states across slave-side impedance, master-side haptic feedback, and gripper execution. This design separates object recognition from direct parameter assignment and provides a unified configuration mechanism for heterogeneous-object teleoperation.
2. **Asynchronous non-blocking strategy deployment.** A multi-rate architecture based on bounded queues and non-blocking polling integrates low-rate visual inference with the nominal 200-Hz supervisory loop. The first threshold-qualified detection triggers a single parameter update, after which trial-level strategy locking prevents repeated intra-trial switching.
3. **Matched human-in-the-loop comparative evaluation.** A 27-block, 135-trial experiment involving three operators and six objects compares complete multi-channel configuration with fixed-parameter, timed manual-selection, semantic-display-only, and impedance-only conditions. The primary C–E comparison quantifies the bundled system-level benefit of coordinating master-side haptic and gripper-execution updates beyond impedance-only reconfiguration.

The proposed framework targets semi-structured teleoperation scenarios in which the platform and workspace remain fixed while object-level interaction requirements vary. Within this scope, it is intended to support consistent multi-channel configuration and efficient task execution during heterogeneous-object teleoperation.

2. Methods and System Implementation

2.1. System Overview and Multi-Rate Architecture

The experimental platform comprised an Omega.7 force-feedback master device, a seven-degree-of-freedom Franka Panda manipulator equipped with a Franka Hand gripper [25], an Intel RealSense D435i camera, and a control computer (Fig. 1). The D435i acquired 424×240 RGB frames at 15 fps (nominally every 66.7 ms); the depth stream was not used in the present implementation. The control computer executed object detection, the nominal 200-Hz supervisory teleoperation loop, strategy-level parameter configuration, haptic rendering, gripper control, and data logging.

RGB frames were acquired at 15 fps (nominally every 66.7 ms) and passed through a single-slot frame queue to limit stale-frame accumulation. YOLO11n inference ran in a separate process and returned object-class labels and confidence scores

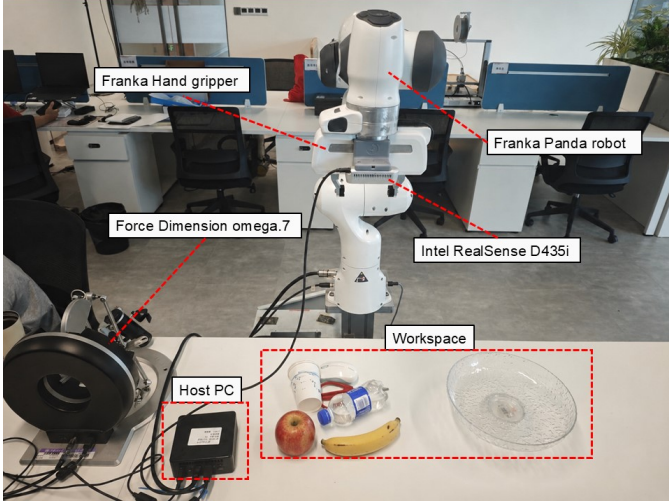


Figure 1: Experimental teleoperation platform showing the Omega.7 master device, a Franka Panda manipulator equipped with a Franka Hand gripper, the Intel RealSense D435i camera, control computer, and object workspace.

through a two-slot result queue. The nominal 200-Hz supervisory loop polled the result queue every 5 ms without blocking for inference, allowing image acquisition, visual inference, and operator-driven teleoperation to proceed concurrently. Per-image processing time was characterized separately using a controlled image test. Table 1 summarizes the operating rates or trigger conditions and associated data flow of the main system modules. Figure 2 illustrates the timing relationships among image acquisition, visual inference, supervisory teleoperation, and strategy-level parameter configuration within the multi-rate asynchronous architecture.

2.2. Teleoperation Mapping and Mechatronic Control Channels

2.2.1. Master-slave motion mapping

Only the translational motion of the Omega.7 was mapped to the Panda end effector. At supervisory step k , the master-device position increment and the desired slave end-effector position were updated as

$$\Delta \mathbf{p}_m(k) = \mathbf{p}_m(k) - \mathbf{p}_m(k-1), \quad (1)$$

$$\mathbf{p}_d(k) = \mathbf{p}_d(k-1) + S \mathbf{C} \Delta \mathbf{p}_m(k), \quad (2)$$

where $\mathbf{p}_m(k) \in \mathbb{R}^3$ denotes the measured Omega.7 position, $\Delta \mathbf{p}_m(k) \in \mathbb{R}^3$ is its incremental displacement, and $\mathbf{p}_d(k) \in \mathbb{R}^3$ denotes the desired Panda end-effector position at supervisory step k . The position-scaling factor was $S = 3.0$, and $\mathbf{C} = \text{diag}(-1, -1, 1)$ defined the axis-direction mapping between the master and slave coordinate conventions. The desired end-effector orientation $\mathbf{R}_d$ was held at a prescribed value throughout each trial; operator-commanded rotational motion was not mapped to the Panda.

2.2.2. Slave-side Cartesian impedance control

For a compact task-space representation, the slave-side Cartesian impedance action was written as

$$\mathbf{w}_c = \mathbf{K}(c)\mathbf{e} + \mathbf{D}(c)\dot{\mathbf{e}}, \quad (3)$$

$$\mathbf{K}(c) = \text{diag}(K_t, K_t, K_t, K_r, K_r, K_r), \quad (4)$$

where $\mathbf{w}_c \in \mathbb{R}^6$ denotes the Cartesian impedance action; $\mathbf{e}, \dot{\mathbf{e}} \in \mathbb{R}^6$ are the pose-error and pose-error-rate vectors; and c denotes the detected object class that indexes the active configuration. The matrices $\mathbf{K}(c)$ and $\mathbf{D}(c)$ denote the associated stiffness and effective damping actions, respectively. The scalar values K_t and K_r specify translational and rotational stiffness, respectively. The error vector $\mathbf{e}$ combines translational error relative to the desired position $\mathbf{p}_d$ and rotational error relative to the fixed desired orientation $\mathbf{R}_d$. With $\mathbf{R}_d$ fixed, K_r set the rotational stiffness against disturbances and deviations from the prescribed orientation. No rotational command was supplied by the operator.

The slave-controller API constructed an internal damping quantity $\mathbf{D}_{\text{API}}(c)$ from the active diagonal stiffness matrix $\mathbf{K}(c)$ and the configuration-dependent dimensionless damping-scaling parameter $\zeta(c)$ according to

$$\mathbf{D}_{\text{API}}(c) = 2\zeta(c)\sqrt{\mathbf{K}(c)}, \quad (5)$$

The square root in Eq. (5) was applied elementwise to the diagonal entries of $\mathbf{K}(c)$. Accordingly, $\mathbf{D}_{\text{API}}(c)$ represents the API's internal damping construction, and $\zeta(c)$ is treated as the dimensionless damping-scaling parameter exposed by the controller. Because no effective Cartesian inertia matrix was identified, $\mathbf{D}_{\text{API}}(c)$ is not reported as a physical damping matrix in SI units, and $\zeta(c)$ is not interpreted as a Cartesian critical-damping ratio.

2.2.3. Master-side haptic rendering

For master-side haptic rendering, the Franka internal state estimator provided a slave-side external-force estimate; its three translational components $\mathbf{F}_{\text{ext}}$, in newtons, were used as inputs to the force-feedback calculation. The baseline force-feedback command along Cartesian axis i was computed as

$$u_{h,i}^{\text{base}} = \text{sgn}(K_f F_{\text{ext},i}) \max(|K_f F_{\text{ext},i}| - d, 0), \quad i \in \{x, y, z\}, \quad (6)$$

Here, $u_{h,i}^{\text{base}}$ denotes the baseline master-side force command along axis i , K_f is a dimensionless force-scaling factor, and d is the per-axis dead-zone threshold in newtons. Equation (6) applies force scaling and a symmetric dead zone independently to each Cartesian component.

In addition to the baseline force-feedback command, a separate cue along the positive z -axis was generated from the Omega.7 gripper angle θ_Ω :

$$\alpha_g = \text{clip}\left(\frac{\theta_{\max} - \theta_\Omega}{\theta_{\max} - \theta_{\min}}, 0, 1\right), \quad (7)$$

$$u_g = \min(\alpha_g G_{\text{cue}} F_{\text{cue,max}}, F_{\text{cue,max}}), \quad u_{h,z} = u_{h,z}^{\text{base}} + u_g. \quad (8)$$

Table 1: Timing, triggers, and data flow of the multi-rate mechatronic system.

Module	Timing or trigger	Data flow
Master input	200 Hz	Omega.7 translational position and button states $\rightarrow$ master-state update.
Slave control	200 Hz	Desired pose and impedance parameters $\rightarrow$ Panda command.
Image acquisition	15 fps (66.7 ms/frame)	D435i RGB stream $\rightarrow$ 424×240 RGB frame.
YOLO11n inference	Asynchronous; per-image processing time characterized separately	Latest available frame $\rightarrow$ object class and confidence score.
Strategy scheduling	First threshold-qualified detection	Object class and confidence $\rightarrow$ target parameter vector $\Theta(s)$.
Haptic rendering	200 Hz	Force-related state $\rightarrow$ Omega.7 force command.
Gripper command	Button event	Gripper input and command parameters $\rightarrow$ Franka Hand command.

Note: YOLO11n inference ran in a separate process and did not block the nominal 200-Hz supervisory loop.

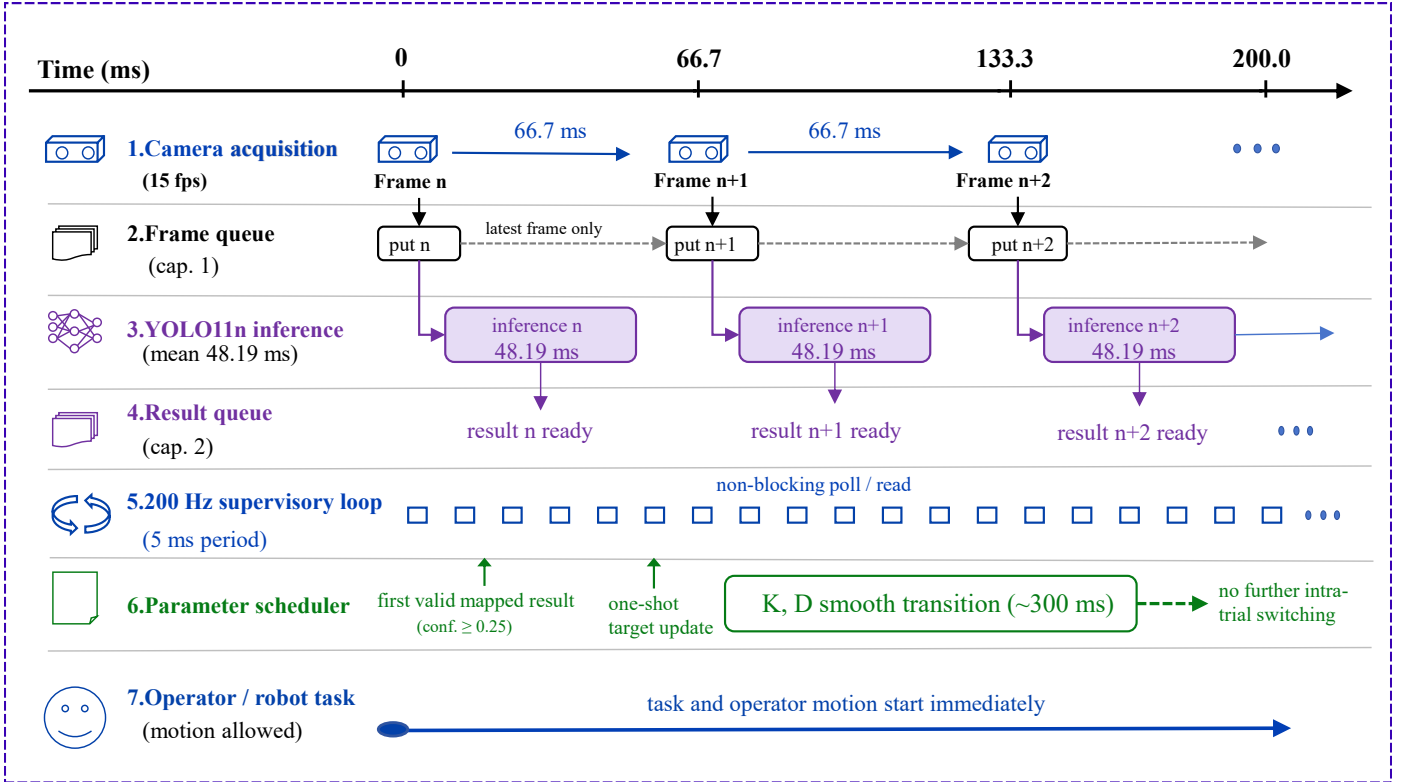


Figure 2: Not-to-scale timing schematic of the asynchronous perception-control architecture. Frames and inference results were exchanged through bounded queues, while the nominal 200-Hz supervisory loop polled the result queue without blocking. The first threshold-qualified result (≥ 0.25 confidence) triggered a single strategy-level update, after which the selected strategy remained locked for the trial; impedance parameters transitioned over approximately 300 ms. The schematic shows nominal module timing; end-to-end trigger latency was not measured.

Here, $\alpha_g \in [0, 1]$ is the normalized Omega.7 gripper-aperture variable, and $\theta_{\min}$ and $\theta_{\max}$ denote the lower and upper gripper-angle bounds used for normalization. The quantities G_{cue} and $F_{\text{cue}, \max}$ denote the cue gain and cue-force limit, respectively, and u_g is the additional positive- z cue added to the baseline command $u_{h,z}^{\text{base}}$. With $G_{\text{cue}} = 0.3$ and $F_{\text{cue}, \max} = 1.0$ N, Eq. (8) reduced to $u_g = 0.3\alpha_g$ N, yielding an additional cue within $[0, 0.3]$ N. The cue was updated during each nominal 200-Hz supervisory cycle and increased as the Omega.7 gripper moved toward the more-open state. It served as a master-side aperture cue based on the Omega.7 gripper aperture. No slave-gripper aperture or independently measured grasp-force signal entered

the cue calculation.

2.2.4. Gripper-execution channel

For grasp execution, v_g was passed as the speed argument to the `grasp()` API, whose other inputs were target width, requested force, `epsilon_inner`, and `epsilon_outer`. The same v_g value was supplied to `move()` during gripper opening and non-grasp position adjustments. The strategy-dependent value F_g specified the requested-force input to `grasp()` in newtons. No independent fingertip-force measurement was used for this variable.

The target-width argument supplied to `grasp()` was fixed at 0 m for all strategies. In the experimental implementation,

ϵ_{inner} and ϵ_{outer} were fixed at 0.005 and 0.080 m, respectively, with 0.080 m corresponding to the maximum gripper aperture. Together with the target-width argument, these tolerances defined the width bounds used by the API-level $\text{grasp}()$ acceptance criterion and remained unchanged across strategies.

After contact and successful entry into the HOLDING state, the $\text{grasp}()$ command was not retransmitted during each supervisory cycle. The gripper maintained the grasp through its internal controller until $\text{stop}()$ was issued, followed by the prescribed release or opening command.

The three parameter channels acted on distinct components of the teleoperation system: slave-side impedance regulated the Panda response to pose error and disturbances, master-side haptic settings mapped the external-force estimate and master-side aperture cue to the Omega.7 force command, and Franka Hand settings specified gripper execution. The proposed framework coordinated these channels through a common strategy-level parameter bundle.

2.3. Object-Conditioned Strategy-Level Parameter Configuration

2.3.1. Object-class-to-strategy and strategy-to-parameter mapping

The framework implemented a two-stage mapping from each detected object class to a manipulation strategy and subsequently from the selected strategy to a multi-channel parameter vector:

$$s = f(c), \quad \Theta(s) = \begin{bmatrix} K_t(s) \\ K_r(s) \\ \zeta(s) \\ K_f(s) \\ d(s) \\ v_g(s) \\ F_g(s) \end{bmatrix}. \quad (9)$$

Here, c denotes the detected object class, and $s \in \mathcal{S}$ is the corresponding manipulation strategy. The strategy set $\mathcal{S}$ comprised the fragility-priority, balanced, and stability-priority strategies. The parameter vector $\Theta(s)$ contains the slave-side impedance parameters $\{K_t, K_r, \zeta\}$, the master-side haptic parameters $\{K_f, d\}$, and the gripper-execution parameters $\{v_g, F_g\}$. For compact notation in the preceding control-channel equations, quantities indexed by c denote values selected through this two-stage mapping.

The three strategies summarized task-relevant interaction priorities involving deformation sensitivity, contact impact, slip risk, pose retention, and grasp stability. The apple and banana were assigned to the fragility-priority strategy because lower stiffness, gripper-closing speed, and requested grasp force were selected to reduce contact impact, squeezing, and deformation. Their smooth surfaces were also considered because they introduced a potential slip risk during grasping and transfer.

The paper cup and bottle were assigned to the balanced strategy because both required a compromise between gentle contact and stable grasping. For the paper cup, excessive force could produce visible deformation, whereas overly low settings

could reduce grasp security. For the bottle, sufficiently stable grasping was required to limit slip during transfer without using unnecessarily aggressive settings.

The mouse and scissors were assigned to the stability-priority strategy because their surface geometry and transfer requirements placed greater emphasis on pose retention and grasp stability. The elongated geometry of the scissors introduced an additional requirement for orientation retention during transfer.

These strategy assignments were defined for the six tested objects, the present teleoperation platform, and the specified grasp-and-transfer task. The corresponding strategy-level parameter settings are listed in Table 2.

2.3.2. One-shot triggering, strategy locking, and parameter transition

At runtime, the first result with confidence ≥ 0.25 retrieved by the supervisory loop triggered a one-shot update of the parameter channels enabled in the active experimental mode. The selected strategy was then locked for the remainder of the trial. The translational stiffness K_t , rotational stiffness K_r , and damping-scaling parameter ζ were transitioned using smoothstep interpolation over a nominal transition duration of $T_s = 0.30$ s. Defining the normalized transition time and smoothstep function as

$$\tau = \text{clip}\left(\frac{t - t_0}{T_s}, 0, 1\right), \quad h(\tau) = 3\tau^2 - 2\tau^3,$$

each parameter $q \in \{K_t, K_r, \zeta\}$ was updated according to

$$q(t) = q_0 + h(\tau)(q_{\text{target}} - q_0). \quad (10)$$

Here, t_0 is the strategy-event time, q_0 is the parameter value at t_0 , and q_{target} is the value specified by the selected strategy. The smoothstep function generated continuous parameter trajectories with zero slope at the beginning and end of the transition. The interpolation was introduced as a command-smoothing mechanism to avoid abrupt changes in the impedance parameters. No formal transient-stability analysis of the complete teleoperation system was performed.

During the transition, the controller recomputed $\mathbf{D}_{\text{API}}$ from the current values of $\mathbf{K}$ and ζ . By contrast, the enabled haptic-interface parameters $\{K_f, d\}$ and gripper-execution parameters $\{v_g, F_g\}$ were updated immediately when the strategy event occurred.

The class-to-strategy mapping was defined manually for the known object classes used in this study. Each recognized class invoked one of the three predefined strategies, providing a bounded and repeatable parameter configuration across the slave robot, haptic interface, and gripper. Online estimation of object physical properties was not included in the mapping. Extending the framework to a new object requires a detector label with reliable recognition and an explicit assignment to one of the predefined strategies before the corresponding parameter vector can be invoked.

Figure 3 summarizes the two-stage object-class-to-strategy-to-parameter mapping, the one-shot strategy event, and the three parameter channels.

Table 2: Strategy-level parameter settings and assigned objects. K_t is reported in N/m, K_r in N m/rad, d in N, and v_g in m/s. F_g denotes the requested-force input to `grasp()` in newtons. The damping-scaling parameter ζ and force-scaling factor K_f are dimensionless.

Strategy	Assigned objects	Slave-side impedance	Master-side haptic feedback	Gripper execution
Fragility-priority	Apple, banana	$K_t = 50$ N/m; $K_r = 5$ N m/rad; $\zeta = 0.8$	$K_f = 0.2$; $d = 0.3$ N	$v_g = 0.02$ m/s; $F_g = 8$ N
Balanced	Paper cup, bottle	$K_t = 150$ N/m; $K_r = 10$ N m/rad; $\zeta = 1.0$	$K_f = 0.5$; $d = 0.4$ N	$v_g = 0.05$ m/s; $F_g = 15$ N
Stability-priority	Mouse, scissors	$K_t = 200$ N/m; $K_r = 13$ N m/rad; $\zeta = 1.2$	$K_f = 0.7$; $d = 0.5$ N	$v_g = 0.10$ m/s; $F_g = 20$ N

Note: F_g is the requested-force input to the Franka Hand `grasp()` API and does not represent an independently measured fingertip force.

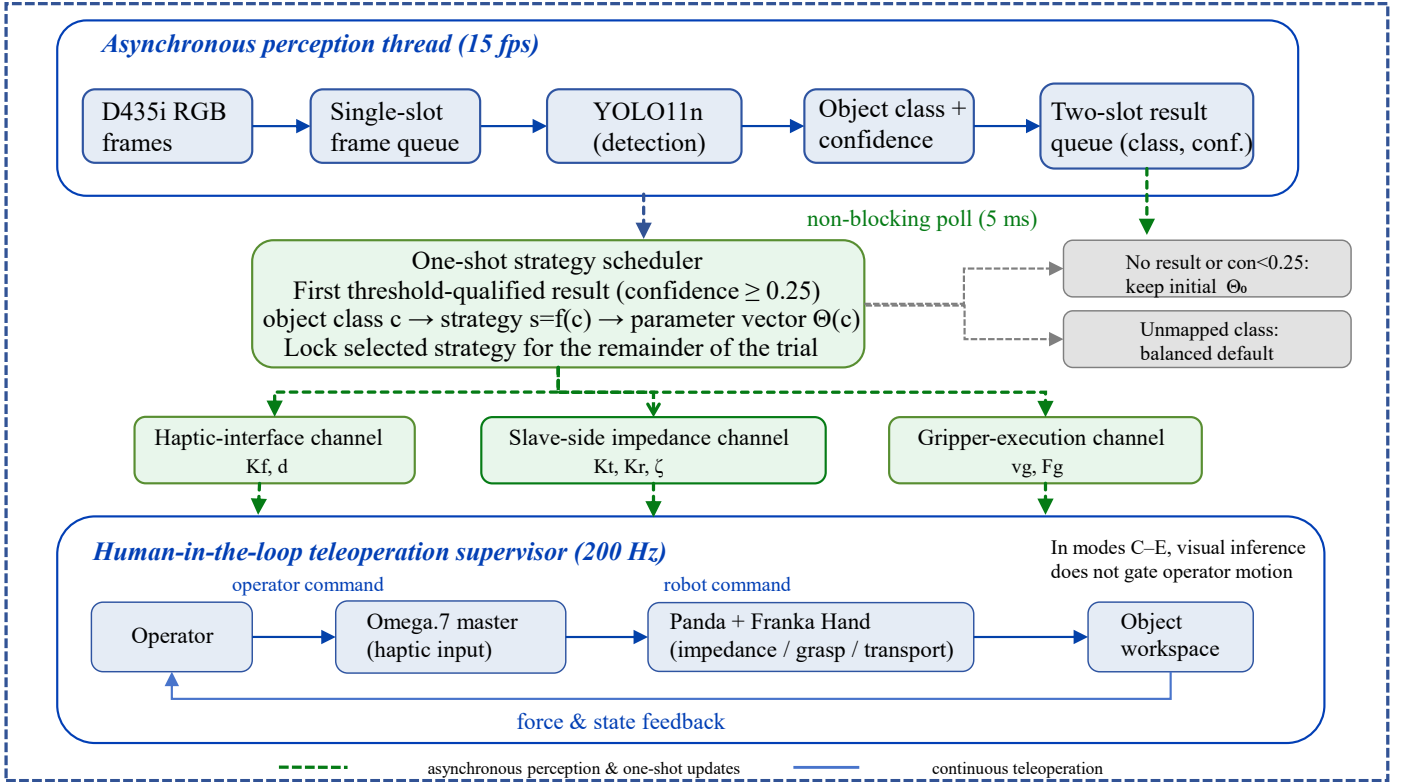


Figure 3: Object-conditioned strategy-level parameter-configuration architecture. D435i RGB frames were processed asynchronously by YOLO11n, and object-class labels and confidence scores were returned through bounded queues. The nominal 200-Hz supervisory loop polled the result queue without blocking. The first result with confidence ≥ 0.25 triggered the object-class-to-strategy mapping and a one-shot update of the enabled slave-side impedance, master-side haptic, and gripper-execution parameter channels. The selected strategy remained locked for the trial. When no threshold-qualified result was available, the initialized parameter state was retained; a detected class without an explicit mapping invoked the balanced default strategy. The continuous operator-robot motion and force-feedback loop remained active throughout task execution.

2.4. Parameter Sets and Initialization

The fixed baseline parameter vector was defined as

$$\Theta_0 = [150 \quad 10 \quad 1.0 \quad 0.5 \quad 0.3 \quad 0.10 \quad 20]^T, \quad (11)$$

Its elements followed the order $[K_t, K_r, \zeta, K_f, d, v_g, F_g]^T$ defined in Eq. (9). The three strategy-level parameter vectors listed in Table 2 were fixed before formal data collection. The baseline vector Θ_0 served as the initialized parameter state and remained active unless it was replaced by a manual or automatic strategy update specified by the experimental mode. In modes with

automatic scheduling, the absence of a result or a confidence value below 0.25 retained Θ_0 , whereas a threshold-qualified class without an explicit mapping invoked the balanced strategy vector.

The three strategy-level operating points were selected through qualitative engineering screening on representative objects before formal data collection. Candidate settings were excluded when they were associated with sustained oscillation, noticeable haptic discomfort, grasp failure, or visible object damage. The final settings listed in Table 2 were fixed before the 135 formal trials. They represent screened operating points

for the present platform, objects, and task, rather than optimized values or material-specific damage limits. The parameter-level design rationale is provided in Supplementary Table S1.

2.5. Mode-Dependent Runtime Procedure

Algorithm 1 summarizes the mode-dependent runtime procedure, including initialization, strategy-event generation, channel-specific parameter updating, strategy locking, data logging, and trial reset. The common supervisory loop executed teleoperation, master-side haptic rendering, gripper control, and data logging in all modes. Visual inference and the visual-semantic display were enabled only in modes C–E, whereas mode B used manual strategy selection after trial timing had begun. Parameter updating was restricted to the channels enabled by the active experimental mode.

Algorithm 1: Mode-dependent runtime procedure for object-conditioned strategy-level parameter configuration

1. Load the initialized parameter state specified by the active experimental mode and start the nominal 200-Hz supervisory loop. Launch the visual-inference process only in modes C–E.
2. Execute teleoperation, master-side haptic rendering, and gripper control continuously within the supervisory loop. In modes C–E, poll the visual-result queue without blocking and update the visual-semantic display using the latest available result.
3. In mode B, begin trial timing before manual object identification and button-based strategy selection. After the operator selects a strategy, generate a manual strategy event and permit robot motion.
4. In modes C and E, while the automatic strategy state remains unlocked, retain the initialized parameter state when no result is available or the confidence is below 0.25. For the first threshold-qualified result, select the strategy explicitly mapped to the detected object class; when no explicit mapping exists, select the balanced default strategy.
5. In mode D, use the visual result only to update the visual-semantic display and retain the initialized parameter vector Θ_0 throughout the trial.
6. Upon a manual or automatic strategy event, update only the parameter channels enabled by the active experimental mode. Transition K_t , K_r , and ζ using smoothstep interpolation over the nominal duration $T_s = 0.30$ s, and update the enabled master-side haptic parameters $\{K_f, d\}$ and gripper-execution parameters $\{v_g, F_g\}$ immediately.
7. For modes with an active strategy update (B, C, and E), lock the selected strategy for the remainder of the trial. In modes C and E, continue the visual-semantic display where enabled, but ignore subsequent detection results for parameter reconfiguration.
8. Continue the prescribed return-and-termination procedure until the robot and gripper reach the predefined terminal state; a recorded failure does not terminate timing early.

9. Log the master-side trajectory, gripper input, active parameter state, and task duration, record the trial outcome, and reset the system for the next trial.

Note: Mode A retained the initialized parameter state without visual inference or strategy updating. Mode B used a manually generated strategy event. Mode C updated all three parameter channels automatically. Mode D displayed visual-semantic information while retaining Θ_0 . Mode E automatically updated only the slave-side impedance channel.

2.6. Fallback Logic and Operational Safeguards

A threshold-qualified misclassification as another known object class invoked the strategy assigned to the predicted class. A threshold-qualified object label without an explicit class-to-strategy mapping invoked the balanced default strategy. When no result was available or confidence was below 0.25, no strategy event was generated and the initialized parameter state was retained. Fallback selection was therefore restricted to retaining the initialized state or invoking one of the three predefined strategy targets; only the parameter channels enabled by the active experimental mode were updated. This restriction prevented the visual module from generating unrestricted parameter targets online, although the suitability of a selected strategy still depended on correct object classification.

In addition to the software fallback logic, operational safeguards included the robot collision-detection function, configured Franka Hand force-command settings, zero-force commands on controller exit, a standardized initial pose, and a manually accessible emergency stop. The Omega.7 was operated using its standard device driver and the hardware configuration used throughout the experiment. The additional master-gripper-aperture cue u_g , applied along the positive z direction, was explicitly limited to 0.3 N. This limit applied only to u_g ; the baseline force-feedback command $u_{h,i}^{\text{base}}$ was transmitted without an additional application-layer saturation. No saturation-status signal was recorded. Accordingly, the retained data characterize the commanded haptic signal but do not establish whether saturation occurred within the driver or device.

3. Experimental Design

3.1. Experimental Objectives, Comparisons, and Outcome Hierarchy

The experiment was designed to evaluate the task-level performance of object-conditioned strategy-level parameter configuration during heterogeneous-object haptic teleoperation. Task completion time was specified as the primary endpoint. The primary planned comparison was between the complete multi-channel configuration mode (C) and the impedance-only reconfiguration mode (E).

Modes C and E used the same visual-semantic display, object-class-to-strategy mapping, and slave-side impedance update. Mode C additionally updated the master-side haptic and gripper-execution parameter channels. The C–E comparison therefore assessed the additional system-level contribution of

jointly configuring these two channels beyond impedance-only reconfiguration. Because the two added channels changed together, the comparison was interpreted at the bundled system level.

Secondary planned contrasts compared mode C with the fixed-parameter baseline (A), the timed manual-selection workflow (B), and the visual-semantic display condition with fixed parameters (D). These contrasts respectively evaluated performance relative to fixed operation, a timed manual-selection workflow, and semantic display alone. Observed task success and Raw NASA-TLX were treated as secondary descriptive outcomes. Master-side trajectory length and pause count were analyzed as exploratory process measures. All analyses were interpreted as repeated-measures evidence for the tested three-operator sample. Population-level human-factors generalization was not the objective of the present experiment.

Table 3 defines the five experimental modes in terms of the information available to the operator, the parameter channels eligible for updating, and the comparison role of each mode. In modes C–E, the visual-semantic display remained active throughout each formal trial. Modes C and D used the same visual-semantic display, while mode C enabled complete multi-channel parameter configuration and mode D retained the initialized parameter state. In mode B, trial timing began before manual object identification and button-based strategy selection. The C–B contrast therefore evaluated the automated workflow against a timed manual-selection workflow, including the associated interaction steps.

3.2. Operators, Objects, and Matched-Block Design

Three operators participated in the main experiment (P01–P03; 23–24 years of age; two male and one female; all right-handed). Because the Omega.7 was positioned on the left side of the platform, all operators used their left hand to control the device in every trial. The device position and operating hand were held constant across operators and experimental modes to standardize the physical interface condition. All operators had received basic teleoperation training and completed a 10–15-min familiarization session immediately before formal data collection. During familiarization, they practiced master-device motion, gripper operation, and the complete task procedure. Written informed consent was obtained from all operators before participation.

The six objects were selected to represent different contact, grasping, and transfer requirements, including deformation sensitivity, smooth or irregular surface geometry, and elongated shape. Each object was assigned to the fragility-priority, balanced, or stability-priority strategy to select the corresponding parameter vector for the present platform and task. The assignments served as operational labels for parameter selection in the specified grasp-and-transfer task and were not intended as a general classification of material properties. Table 4 summarizes the approximate physical profiles, assigned strategies, and principal manipulation considerations for the six objects. The paper cup and scissors provide contrasting examples of deformation-sensitive handling and orientation-sensitive transfer, respectively.

The main experiment comprised 27 matched task blocks. Each block was defined by a unique combination of operator, object, and repetition index and contained one trial under each of modes A–E. Accordingly, 27 blocks $\times$ 5 modes per block yielded 135 trials. The five mode-specific observations within each block therefore formed the matched set for the completion-time analysis.

For each operator and strategy category, the three matched blocks were allocated between the two assigned objects in a 2:1 ratio. The same object was used for all five mode trials within a block, thereby preserving object-matched comparisons. The 2:1 allocation between the two objects was reversed across operators. This schedule yielded five blocks for one object and four blocks for the other within each strategy, as summarized in Table 5. The object allocation was fixed in the experimental schedule before formal data collection.

For the nine operator $\times$ strategy groups, the five-mode orders were preassigned using nine different sequences. The schedule varied the position of each mode across the nine groups, although it did not implement a complete Latin-square counterbalance. Across the nine sequences, mode C appeared in the fourth position for three groups, while mode E appeared in the fifth position for three groups. The complete mode-order sequences are provided in Supplementary Table S2. With three operators, the effects of mode order, object allocation, and operator could not be estimated independently. The C–E comparison was therefore treated as a paired contrast within the observed sample.

3.3. Task Procedure and Outcome Measures

Before each trial, the test object was manually placed within a predefined workspace region and aligned to the prescribed orientation as closely as practicable. Small trial-to-trial variations in object position and orientation remained after manual placement; the same placement protocol was applied across all five modes. Only translational master-device motion was mapped to the Panda end effector, while the desired end-effector orientation remained fixed throughout the trial.

Each trial comprised six stages: reset, approach, grasp, transfer, release, and termination. In modes C–E, camera acquisition, visual inference, and teleoperation began concurrently at task initiation. Operator-driven motion was permitted while visual inference proceeded asynchronously. In mode B, trial timing began before manual object identification and button-based strategy selection. Robot motion was enabled after the operator completed the selection step. The end of the manual selection step was not recorded with a separate timestamp.

A trial was classified as successful when the object was grasped, transferred, and placed without a drop, observable slip, or visible damage. A drop was defined as complete separation of the object from the gripper. Slip was defined as visible relative motion between the object and gripper during grasping or transfer. Damage was defined as a clear indentation, crushing deformation, or other visible object damage. A designated researcher applied the predefined outcome criteria during each trial. Outcome labels were assigned from direct observation dur-

Table 3: Experimental modes, information conditions, channel-update logic, and comparison roles.

Mode	Operator information / selection	Slave-side impedance	Master-side haptic feedback	Gripper execution	Comparison role
A: Fixed-Parameter Baseline	No visual-semantic display or strategy selection	Fixed	Fixed	Fixed	Fixed-parameter reference condition.
B: Timed Manual-Selection Workflow	No visual-semantic display; operator identifies the object and selects a complete strategy after timing begins.	Manual	Manual	Manual	Timed manual-workflow reference.
C: Complete Multi-Channel Configuration	Object class, selected strategy, confidence score, and color-coded bounding box	Automatic	Automatic	Automatic	Complete multi-channel condition; primary condition in the C-E comparison.
D: Visual-Semantic Display with Fixed Parameters	Same visual-semantic display as mode C	Fixed	Fixed	Fixed	Semantic-display control condition.
E: Impedance-Only Re-configuration	Same visual-semantic display as mode C	Automatic	Fixed	Fixed	Impedance-only reference for the primary C-E comparison.

Note: Automatic denotes a one-shot update triggered by the first detection result with confidence ≥ 0.25 . Manual denotes a one-shot complete-strategy update after operator selection, with identification and selection included in trial timing. Fixed denotes retention of the initialized parameter state. In modes C–E, no result or confidence below 0.25 produced no parameter update. A threshold-qualified class without an explicit mapping invoked the balanced default strategy; mode D displayed the visual-semantic output while retaining Θ_0 . Visual inference was not used in modes A or B. The C–E comparison was interpreted at the bundled system level because the master-side haptic and gripper-execution channels changed together.

Table 4: Test objects, assigned strategies, approximate physical profiles, and principal manipulation considerations.

Object	Assigned strategy	Approximate physical profile	Principal manipulation consideration
Apple	Fragility-priority	~200 g; smooth surface; diameter 70–80 mm	Contact impact and slip; gentle grasping required.
Banana	Fragility-priority	~120 g; smooth surface; 20×160 mm	Compression-related deformation and slip.
Paper cup	Balanced	~5 g; paper; diameter 55 mm; height 60 mm	High deformation sensitivity with a concurrent stability requirement.
Bottle	Balanced	~30 g; smooth plastic; diameter 65 mm; height 100 mm	Transfer slip and the balance between gentle contact and grasp stability.
Mouse	Stability-priority	~100 g; smooth plastic; $65 \times 100 \times 35$ mm	Rigid, irregular geometry and potential transfer slip.
Scissors	Stability-priority	~150 g; metal and plastic; $50 \times 130 \times 15$ mm	Elongated rigid geometry and a high orientation-retention requirement.

Table 5: Distribution of matched task blocks and trials across strategies and objects.

Object	Blocks	Trials
<i>Fragility-priority</i>		
Apple	4	20
Banana	5	25
<i>Balanced</i>		
Paper cup	5	25
Bottle	4	20
<i>Stability-priority</i>		
Mouse	5	25
Scissors	4	20
Total (six objects)	27	135

Note: Each matched block contains five trials—one under each experimental mode—so the numbers of trials equal five times the corresponding numbers of blocks.

ing the experiment; systematic video recording and independent blinded re-evaluation were not performed.

The occurrence of a failure did not terminate trial timing. Regrasping after a drop was not permitted, and the trial was

recorded as a failure. The operator then completed the prescribed return-and-termination procedure until the robot and gripper reached the predefined terminal state. Successful and failed trials therefore used the same predefined timing endpoint. The retained dataset included the master-side trajectory, gripper input, active parameter state, task duration, and outcome label. The trial logs did not retain event-level object-class labels, confidence scores, strategy-trigger timestamps, or synchronized trigger-to-contact timestamps. Consequently, the retained data cannot retrospectively determine the trigger timing relative to contact for every trial.

Task completion time was measured from task initiation to the common predefined terminal state and was retained for both successful and failed trials. Observed success rate was calculated as the number of successful trials divided by the total number of trials in each mode. It was reported descriptively because the outcome labels were assigned from direct observation without independent re-evaluation.

Subjective workload was assessed using the Raw NASA-TLX questionnaire [26]. Each of the six dimensions was rated on a 0–100 scale, and the unweighted arithmetic mean of the six ratings was used as the Raw NASA-TLX score. Each operator completed one questionnaire immediately after completing

the three trials associated with a given strategy $\times$ mode condition. The three trials combined the two objects assigned to that strategy according to the operator-specific 2:1 allocation. The workload dataset therefore comprised 3 operators $\times$ 3 strategies $\times$ 5 modes = 45 questionnaires, corresponding to nine questionnaire units per mode. This questionnaire structure supported descriptive comparisons at the mode and strategy levels. Object-specific workload was not measured separately.

Master-side trajectory length and pause count were analyzed as exploratory process measures. Visual performance was characterized separately in a controlled image test using object-class accuracy, class-to-strategy mapping accuracy, confidence score, and per-image processing time. Before analysis, a pause was operationally defined as a continuous interval during which the master-device translational speed remained below 0.005 m/s for at least 0.30 s. Master-device speed was obtained by differentiating the raw master-side position trajectories, which were sampled at approximately 200 Hz; pause intervals were then identified using the predefined speed and duration thresholds.

3.4. Statistical Analysis

Completion time across the five modes was analyzed using a Friedman test, treating the 27 operator-nested matched task blocks as the repeated-measures units. When the omnibus Friedman test was significant, the four planned paired contrasts comparing mode C with modes A, B, D, and E were evaluated using Wilcoxon signed-rank tests with Holm adjustment for multiple comparisons.

For each matched block b , the primary C–E completion-time difference was defined as

$$\Delta T_b = T_{E,b} - T_{C,b}, \quad (12)$$

so that a positive value favored mode C. The analysis reported the mean paired difference $\bar{\Delta T}$, the relative reduction with mode E as the reference, and a matched-block percentile bootstrap 95% confidence interval. The interval was calculated from 10,000 resamples with replacement of the 27 matched-block differences. Operator-specific mean paired differences were reported to assess directional consistency across operators. A leave-one-operator-out sensitivity analysis was additionally performed to evaluate whether the overall C–E difference was dominated by any single operator.

Observed success outcomes were summarized as counts and percentages for each mode, and paired C–E outcomes were cross-tabulated within the matched blocks. Given the small number of discordant C–E pairs, the exact two-sided McNemar test was treated as a supplementary analysis. Raw NASA-TLX was summarized descriptively across the nine operator $\times$ strategy questionnaire units per mode; no inferential test or bootstrap confidence interval was calculated.

Master-side trajectory length and pause count were treated as exploratory process measures. For the C–E trajectory-length comparison, the mean paired difference and a matched-block percentile bootstrap 95% confidence interval were reported using the same 10,000-resample procedure. Pause counts were summarized descriptively; no formal hypothesis test was prespecified

or performed. Non-binary outcomes were summarized using median [interquartile range (IQR)] and mean $\pm$ standard deviation (SD), as appropriate; categorical success outcomes were reported as counts and percentages. Because the 27 matched blocks were nested within only three operators, the resulting p -values and matched-block bootstrap intervals were interpreted conditionally for the observed repeated-measures sample and were not treated as estimates of population-level participant effects.

4. Results

4.1. Controlled Visual Test and Supervisory-Loop Timing

The separate controlled closed-set visual test used 180 photographs, with 30 images per object class, acquired under fixed viewpoint, background, and lighting conditions. All 180 images were classified correctly and mapped to the intended manipulation strategy. The mean confidence score was 0.853, and the mean per-image processing time was 48.19 ms (Fig. 4). Class-wise results and the confusion matrix are provided in Supplementary Table S3 and Supplementary Fig. S1.

These results characterize closed-set recognition and strategy mapping for the six tested object instances under the specified controlled imaging conditions. They do not quantify the correctness of online strategy triggering, trigger-to-contact timing, or performance on untested objects.

The median logged duration of the nominal 200-Hz supervisory cycle was 5.07 ms.

4.2. Overall Results Across the Five Modes

Across the five modes, mode C showed the lowest median completion time, the highest observed success rate, the lowest median Raw NASA-TLX score, and the lowest median master-side trajectory length (Table 6 and Figs. 5–6). Using the trial-level data in Supplementary Table S4, the mean completion time was 19.28 ± 1.30 s in mode C, compared with 21.42 ± 1.58 s in mode A, 21.01 ± 1.61 s in mode B, 20.91 ± 1.10 s in mode D, and 21.07 ± 1.56 s in mode E. With each comparison mode used as the reference, the corresponding mean reductions for mode C were 10.0%, 8.2%, 7.8%, and 8.5%, respectively. Consistent with the predefined outcome hierarchy, inferential analyses focused on task completion time as the primary endpoint.

Using the 27 operator-nested matched task blocks as repeated-measures units, the Friedman test indicated an overall difference in completion time across the five modes ($\chi^2(4) = 30.904$, $p < 0.001$). Holm-adjusted Wilcoxon signed-rank tests indicated lower completion times in mode C than in modes A, B, D, and E, with all adjusted p -values below 0.01. These comparisons were interpreted as conditional repeated-measures evidence for the observed three-operator sample. Descriptively, mode C yielded the lowest Raw NASA-TLX score within each of the nine operator $\times$ strategy questionnaire units.

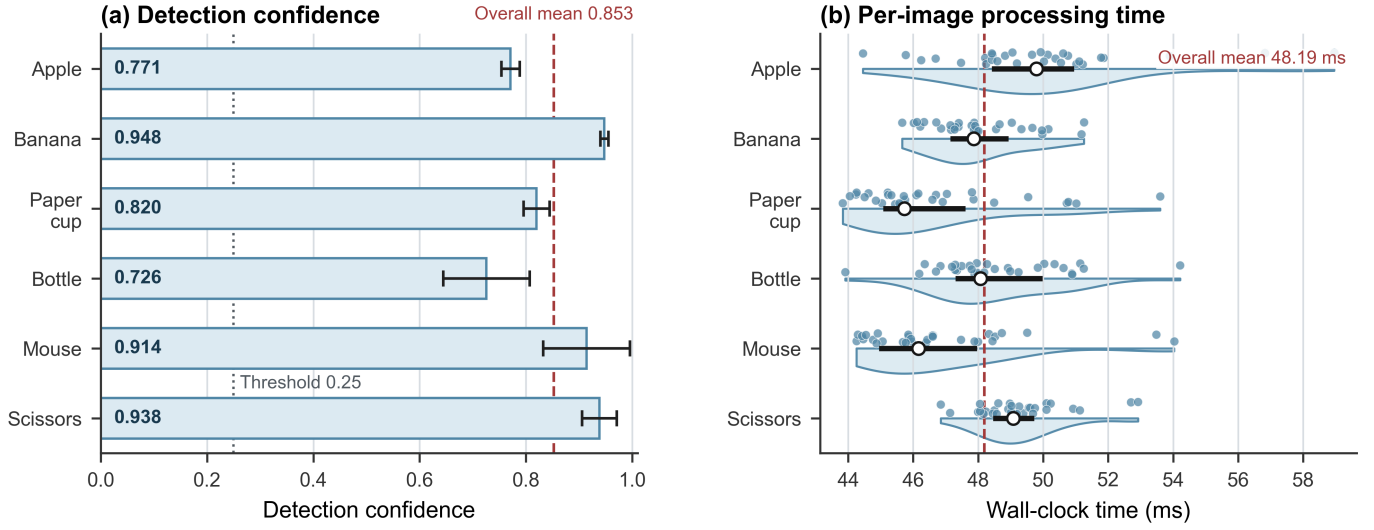


Figure 4: Confidence scores and per-image processing time in the controlled closed-set visual test ($n = 30$ images per object class). Bars indicate mean $\pm$ SD; half violins, points, segments, and open circles indicate distributions, individual observations, IQRs, and medians, respectively. Reference lines indicate the confidence threshold of 0.25 and the overall means for confidence (0.853) and processing time (48.19 ms). The reported processing time does not represent end-to-end task-trigger latency.

Table 6: Descriptive task-performance and workload results across the five experimental modes.

Mode	Task completion time (s)	Master-side trajectory length (m)	Observed success, n/N (%)	Raw NASA-TLX
A	21.18 [20.62, 22.08]	0.757 [0.693, 0.816]	22/27 (81.5%)	62.50 [59.67, 64.50]
B	20.89 [20.12, 21.83]	0.787 [0.721, 0.861]	21/27 (77.8%)	56.17 [55.00, 59.33]
C	19.57 [18.41, 20.05]	0.697 [0.660, 0.769]	26/27 (96.3%)	48.67 [47.67, 51.83]
D	20.79 [20.32, 21.16]	0.722 [0.678, 0.768]	24/27 (88.9%)	60.33 [57.33, 62.50]
E	20.73 [19.95, 22.25]	0.732 [0.678, 0.799]	24/27 (88.9%)	53.67 [51.83, 57.83]

Note: Continuous outcomes are reported as median [IQR], whereas observed success is reported as successful trials/total trials (%). Task completion time, master-side trajectory length, and observed success use $n = 27$ trial-level observations per mode. Raw NASA-TLX uses $n = 9$ operator $\times$ strategy questionnaire units per mode. Mode definitions are provided in Table 3, and the trial-level task-performance data are provided in Supplementary Table S4.

Table 7: Primary within-sample comparison of complete multi-channel configuration and impedance-only reconfiguration.

Outcome	Mode C, median [IQR]	Mode E, median [IQR]	Mean paired difference Δ (E-C) [95% CI]	Operator-level direction
Task completion time (s)	19.57 [18.41, 20.05]	20.73 [19.95, 22.25]	1.79 [1.10, 2.51]	3/3 favored C
Master-side trajectory length (m)	0.697 [0.660, 0.769]	0.732 [0.678, 0.799]	0.024 [-0.014, 0.059]	Mixed
Raw NASA-TLX	48.67 [47.67, 51.83]	53.67 [51.83, 57.83]	4.87 [—]	3/3 favored C

Note: Differences were defined as mode E minus mode C, so positive Δ values favor mode C for all three outcomes. Task completion time and master-side trajectory length use $n = 27$ matched task blocks. Raw NASA-TLX uses $n = 9$ paired operator $\times$ strategy questionnaire units per mode. Confidence intervals for completion time and trajectory length are matched-block bootstrap 95% intervals based on 10,000 resamples and are conditional on the observed block structure. No bootstrap confidence interval was calculated for Raw NASA-TLX.

4.3. Primary C–E Comparison and Exploratory Process Measures

The primary paired comparison between modes C and E is summarized in Table 7 and Fig. 7.

Median task completion time was 19.57 s in mode C and 20.73 s in mode E. Defining the block-level difference as $\Delta T = T_E - T_C$, the mean paired difference was 1.79 s, with a matched-

block bootstrap 95% confidence interval of 1.10–2.51 s. Relative to the mean completion time in mode E, this difference corresponded to an 8.5% reduction in mode C. Mode C had a lower completion time than mode E in 22 of the 27 matched task blocks.

In a post hoc sensitivity analysis, the C–E completion-time comparison was repeated using the 23 matched blocks in which

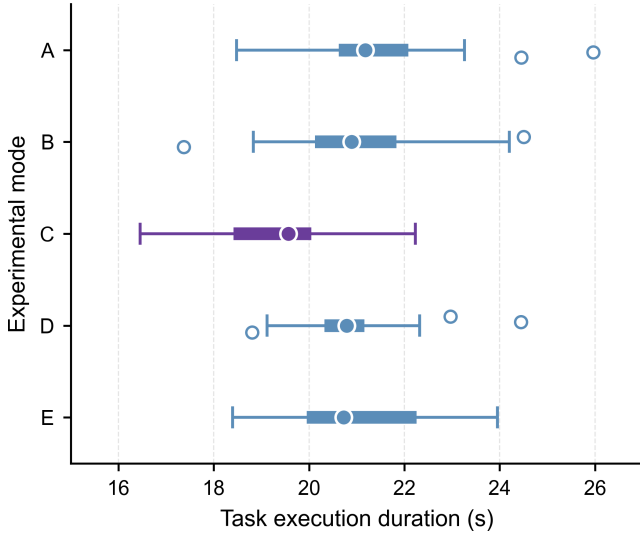


Figure 5: Task completion time across the five experimental modes ($n = 27$ matched task blocks per mode). Filled circles, thick horizontal segments, whiskers, and open circles indicate medians, IQRs, $1.5 \times \text{IQR}$ ranges, and outliers, respectively. This descriptive panel displays marginal mode distributions; the paired C–E observations are shown in Fig. 7.

both mode trials were successful. The mean paired difference in this subset was 1.80 s, and mode C was faster in 19 of the 23 blocks. The estimated difference was similar in magnitude to the 1.79-s primary estimate.

For the paired success outcomes, 23 matched blocks were successful in both modes, three were successful in mode C only, one was successful in mode E only, and none failed in both modes. The exact two-sided McNemar test yielded $p = 0.625$ and, given only four discordant pairs, did not provide clear evidence of a mode-related difference in observed success. Accordingly, the observed success-rate difference was treated as supplementary descriptive evidence.

Median master-side trajectory length was 0.697 m in mode C and 0.732 m in mode E. With $\Delta L = L_E - L_C$, the mean paired trajectory-length difference was 0.024 m, with a matched-block bootstrap 95% confidence interval of -0.014 to 0.059 m. Because the interval included zero, the data did not provide clear evidence of a C–E difference in master-side trajectory length. Pause counts were lower on average in mode C, although the two modes had the same median pause count.

Using the predefined pause criterion in Section 3.3, mode C had a median [IQR] pause count of 3 [2, 3.5] and a mean of 2.74 ± 1.23 , whereas mode E had a median pause count of 3 and a mean of 3.41 ± 1.67 . The mean pause count was lower in mode C within each of the three manipulation strategies, while the overall median remained 3 in both modes. Pause count was treated descriptively, and no formal hypothesis test was performed. The robustness of the result to alternative speed or duration thresholds was not evaluated.

4.4. Observed Failure Outcomes

Across the 135 trials, 18 were classified as failures: 5, 6, 1, 3, and 3 in modes A–E, respectively. The observed failure

outcomes involved object drops, observable slip, or visible damage according to the predefined criteria in Section 3.3. Table 8 summarizes the mode-level failure counts and representative trial observations.

Among the five modes, mode C had the lowest observed failure count, with 1 failure in 27 trials.

4.5. Directional Consistency Across Operators and Objects

The mean paired C–E completion-time difference favored mode C for each of the three operators and each of the six tested objects.

At the operator level, the mean paired differences $\Delta T = T_E - T_C$ were 1.66, 2.56, and 1.16 s for P01, P02, and P03, respectively. The corresponding operator-specific median [IQR] differences were 1.91 [0.60, 3.14], 2.63 [2.19, 3.34], and 1.60 [-0.53, 1.73] s, respectively. Mode C had a shorter completion time in 7/9, 9/9, and 6/9 matched task blocks for P01, P02, and P03, respectively. In the leave-one-operator-out sensitivity analysis, P01, P02, and P03 were excluded in turn. The mean paired differences in the remaining blocks were 1.86, 1.41, and 2.11 s, respectively, and all remained positive.

At the object level, mode C had the lowest mean task completion time among the five modes for each of the six tested objects (Supplementary Table S5). Relative to mode E, the mean completion-time reduction in mode C ranged from 3.3% for the bottle to 13.2% for the scissors. The mean C–E difference favored mode C for all six objects, while the magnitude varied across objects.

5. Discussion

5.1. Principal Findings and Interpretation of the Five-Mode Design

The five-mode evaluation produced one primary finding and several supporting descriptive observations. Mode C had the shortest task completion time among the five tested modes. In the predefined primary C–E comparison, complete multi-channel configuration was associated with a 1.79-s (8.5%) lower mean task completion time, with a matched-block bootstrap 95% confidence interval of 1.10–2.51 s. The mean C–E difference favored mode C for all three operators and all six tested objects. Observed success rate and Raw NASA-TLX also showed descriptive patterns favoring mode C, while the C–E trajectory-length confidence interval included zero. Together, these findings support a within-sample task-level advantage of complete multi-channel configuration under the tested conditions.

The five-mode design helps locate the level at which this advantage should be interpreted. The C–A contrast benchmarks the complete configuration against fixed-parameter operation. The C–B contrast compares automatic strategy selection with a timed manual-selection workflow that includes object identification and button-based strategy selection. Modes C–E used the same visual-semantic display throughout each trial. Mode D displayed the same object class, mapped strategy, confidence score, and bounding box as mode C while retaining the initialized parameter state. The C–D result therefore supports an

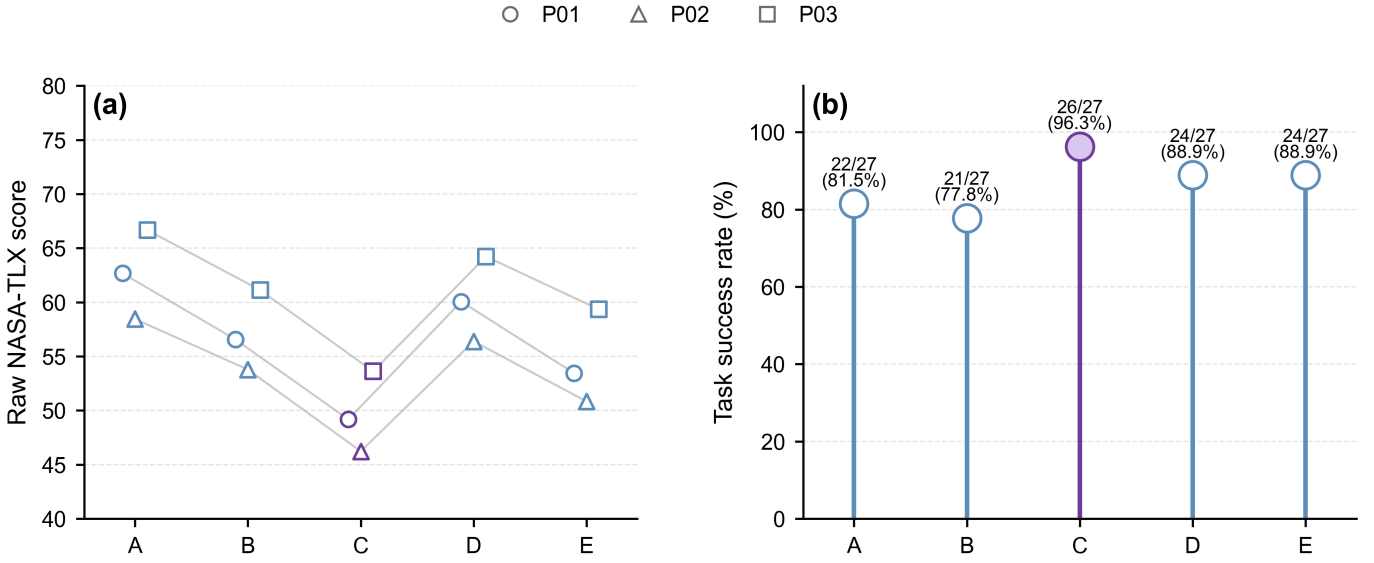


Figure 6: Descriptive workload and observed task-success results across the five experimental modes. (a) Raw NASA-TLX scores: connected markers show each operator’s mean across the three strategy-level questionnaire units within each mode ($n = 9$ operator $\times$ strategy units per mode), and marker shape identifies the operator. (b) Number of successful trials out of 27 trials per mode. Both panels are descriptive; no confidence intervals or primary inferential tests are shown.

Table 8: Observed failure counts and representative trial observations by experimental mode.

Mode	Observed failures, n/N	Representative trial observations
A	5/27	Paper-cup deformation and difficulty maintaining the scissors’ pose were noted in failed trials.
B	6/27	No consistent failure pattern could be assigned from the retained trial records.
C	1/27	Observable mouse slip during transfer.
D	3/27	Object drops or loss of grasp after grasp establishment.
E	3/27	Observed drops, slip, or visible damage during grasp establishment or transfer.

Note: Mode definitions are provided in Table 3. Failure labels were assigned during the experiment using the predefined criteria in Section 3.3. The listed observations are representative rather than exhaustive. Systematic video recording and independent blinded re-evaluation were not performed; the observations should not be interpreted as confirmed root-cause classifications.

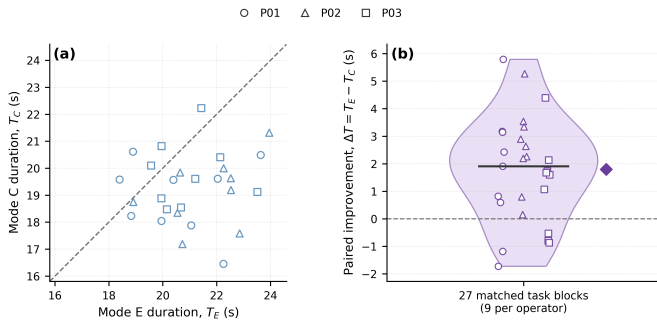


Figure 7: Paired task completion times for modes C and E across 27 matched task blocks. Points below the identity line indicate shorter completion time in mode C. For the paired difference $\Delta T = T_E - T_C$, positive values favor mode C. The violin, horizontal segment, and diamond show the distribution, median, and mean of the paired differences, respectively. The mean paired difference was 1.79 s (matched-block bootstrap 95% CI, 1.10–2.51 s), and 22 of 27 blocks favored mode C.

additional task-level contribution from parameter configuration beyond semantic display alone.

The primary C–E contrast provides the most specific system-level comparison. Both modes used the same object-class-to-strategy mapping, visual-semantic display, and slave-side impedance update, while mode C additionally configured the master-side haptic and gripper-execution channels. The C–E result therefore supports an additional system-level contribution from coordinating the master-side haptic and gripper-execution channels with slave-side impedance. Because the two added channels changed together, this comparison is interpreted at the bundled system level.

The C–E trajectory-length estimate was small, and its bootstrap confidence interval included zero. Mode C showed a lower mean pause count, although the two modes had the same median pause count. No confirmatory analysis was performed for the pause measure. Taken together, the process measures suggest that the time reduction may be associated with fewer corrective interruptions in some trials, while the aggregate geometric path

length changed little. This process-level interpretation remains exploratory.

5.2. Asynchronous Perception–Control Integration

Because semantic output was used for parameter reconfiguration only when generating the one-shot strategy event, the perception process did not need to operate at the nominal 200-Hz supervisory rate. In the implemented multi-rate architecture, 15-fps image acquisition and independent visual inference communicated with the nominal 200-Hz supervisory loop through bounded queues. The single-slot frame queue limited stale-frame accumulation, while non-blocking polling allowed the supervisory loop to retrieve available inference results without waiting for the inference process to complete.

The mean per-image processing time of 48.19 ms was substantially longer than one nominal supervisory period, whereas the median logged supervisory-cycle duration was 5.07 ms. This difference in timescales highlights the role of asynchronous decoupling in the present architecture. Accordingly, the implemented architecture was designed for one-shot strategy configuration followed by strategy locking; continuous visual servoing was outside the implemented control structure.

The implementation supports the feasibility of asynchronous, non-blocking integration of low-rate semantic perception into the supervisory software architecture. The reported timing results characterize software-level integration and typical module timing. Hard real-time guarantees and end-to-end trigger-to-contact timing remain uncharacterized.

5.3. Variation Across Operators and Objects

The positive mean C–E completion-time difference was observed for all three operators and persisted in the leave-one-operator-out analyses. Mode C also had a lower mean task completion time than mode E for each of the six tested objects, with descriptive reductions ranging from 3.3% to 13.2%. Together, these directional checks show that the aggregate C–E difference was distributed across operators and objects in the observed dataset. The operator- and object-level summaries are shown in Fig. 8.

The magnitude of the C–E difference nevertheless varied across objects, with smaller reductions for the bottle and banana and larger reductions for the paper cup and scissors. This pattern may be related to differences in grasp-pose requirements, deformation sensitivity, and the need to retain object orientation during transfer. The larger reductions for the paper cup and scissors are compatible with greater value from complete multi-channel configuration when deformation management or orientation retention becomes more demanding. Each object-level summary was based on four or five matched task blocks and was therefore treated descriptively. Targeted experiments with larger per-object samples and additional objects are needed to determine whether the observed variation follows specific object or task properties.

5.4. Relation to Previous Work and Mechatronic Design Implications

Prior tele-impedance and variable-impedance studies have adapted slave-side impedance using human-state information, contact-force information, demonstrations, or task phase [6–13]; vision-based methods have mapped object attributes, geometry, or vision–language representations to stiffness or impedance settings [14, 15, 17, 18]. Related work on automatic switching, shared control, multi-stiffness interfaces, and teleoperated grasping has addressed mode selection, control-authority allocation, and gripper impedance based on tactile deformation or grip-safety information [16, 19, 20, 24, 27, 28].

The present framework complements these directions by using a single object-conditioned strategy event to coordinate a strategy-level operating state across slave-side impedance, master-side haptic feedback, and gripper execution. The principal contribution is therefore a cross-subsystem, strategy-level configuration architecture together with a system-level evaluation of the resulting multi-channel coordination. At the software-architecture level, bounded queues, non-blocking polling, one-shot triggering, and strategy locking decouple low-rate semantic perception from the nominal 200-Hz supervisory execution path and prevent repeated parameter switching within a trial.

This design principle may be useful in semi-structured applications such as flexible sorting, remote maintenance, and dismantling, where a limited set of recurring object or task classes can be associated with predefined and experimentally screened operating strategies. Extending the framework to larger class sets would require systematic maintenance of the object-class-to-strategy mapping and may require a richer strategy space than the three levels evaluated here. Tactile or contact-state sensing could be added as a post-contact verification layer, enabling corrective updates after contact without requiring the visual process to operate within the nominal 200-Hz supervisory execution path.

5.5. Scope and Future Extensions

The formal-trial logs retained task duration, master-side trajectory, gripper input, active parameter state, and outcome labels, but not event-level object-class labels, confidence scores, complete visual-event histories, or synchronized strategy-event/contact timestamps. Consequently, online trigger correctness, fallback invocation, trigger-to-contact intervals, and completion of the strategy-level update before contact could not be unambiguously reconstructed retrospectively. The controlled 180-image test characterizes closed-set recognition under fixed imaging conditions; trial-level verification of contact-preparatory timing requires synchronized visual, strategy-event, and contact logging.

The experimental evidence was obtained from repeated measurements involving three operators, six known objects, translational master motion, a fixed end-effector orientation, prescribed object placement, and preassigned mode sequences without complete counterbalancing. The current implementation used a manually defined object-class-to-strategy mapping, so extension to a new object requires a detector label and an explicit strategy

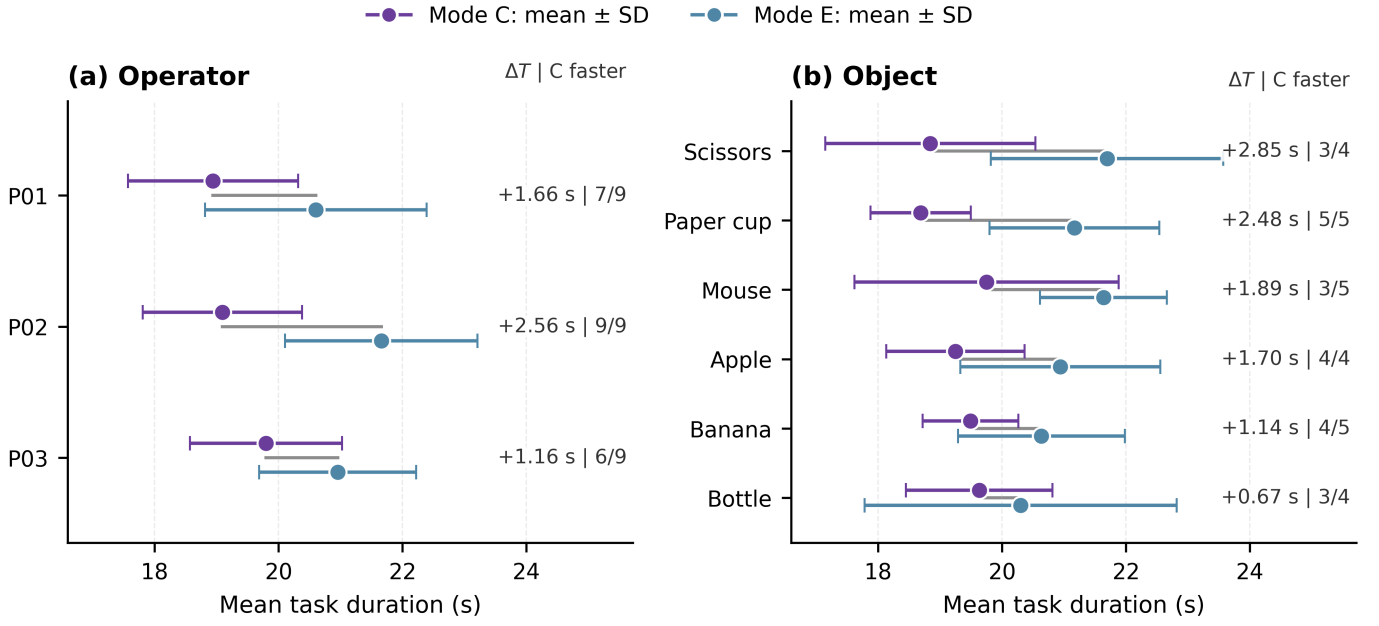


Figure 8: Operator- and object-level summaries of task completion time for modes C and E. (a) Mean task completion time by operator ($n = 9$ matched task blocks per operator). (b) Mean task completion time by object ($n = 4$ or 5 matched task blocks per object). Markers and horizontal error bars indicate mean $\pm$ SD for each mode. Right-side annotations report the mean paired difference ($\Delta T = T_E - T_C$) and the number of blocks in which mode C was faster divided by the total number of blocks. Positive ΔT values favor mode C. Error bars show within-group variability and are not confidence intervals for the paired differences.

assignment. The findings therefore characterize the present platform and task within the tested participant sample. Broader evaluation should incorporate larger and more diverse operator cohorts, fully counterbalanced mode orders, and expanded motion, orientation, and placement conditions while preserving matched comparisons across experimental modes.

The reported parameter vectors were selected through qualitative engineering screening and should be interpreted as reproducible operating points for the present platform, objects, and task. Because the C–E contrast changed the master-side haptic and gripper-execution channels together, separate channel-level ablations are needed to resolve their individual and combined contributions. Outcome labels were based on direct observation, and the retained records did not include independent video-based re-rating or direct contact-quality and corrective-action measurements. Similarly, the retained haptic data characterize the commanded signal but not realized device output or saturation within the driver or device. Future experiments should combine synchronized visual, strategy-event, and contact logging with separate channel-level ablations, independent video-based outcome assessment, direct contact and realized haptic-output measurements, and systematic parameter optimization. An evidence-boundary overview is provided in Supplementary Table S6.

6. Conclusion

This study presented an object-conditioned strategy-level multi-channel parameter-configuration framework for haptic teleoperation of heterogeneous objects. Each detected object

class was mapped to one of three interpretable manipulation strategies, which jointly specified the slave-side impedance, master-side haptic feedback, and gripper-execution parameters. Low-rate visual perception was integrated asynchronously with the nominal 200-Hz supervisory loop through a one-shot strategy event, and the selected strategy remained locked for the remainder of the trial.

Across 27 matched task blocks comprising 135 physical trials, complete multi-channel configuration yielded a 1.79-s lower mean task completion time than impedance-only reconfiguration, corresponding to an 8.5% reduction (matched-block bootstrap 95% CI, 1.10–2.51 s). The mean C–E completion-time difference favored complete multi-channel configuration for all three operators and all six tested objects. This mode also showed the highest observed success rate and the lowest median Raw NASA-TLX score; both outcomes were interpreted descriptively. The C–E confidence interval for master-side trajectory length included zero.

These findings support a within-sample system-level benefit of coordinating slave-side impedance, master-side haptic feedback, and gripper execution beyond impedance-only reconfiguration under the tested conditions. Because the master-side haptic and gripper-execution channels changed together, their individual contributions remain to be isolated. Overall, the framework provides an interpretable system-design approach to object-conditioned multi-channel configuration for semi-structured haptic teleoperation tasks.

Supplementary Material

Supplementary Tables S1–S6 and Supplementary Fig. S1 are provided in a separate supplementary file.

Declarations

Ethics statement

The school confirmed that formal ethics committee review was not required for this low-risk, non-invasive study. All participants provided written informed consent before participation. The experimental procedure was conducted in accordance with relevant institutional guidelines; no personally identifiable information was collected, and all data were anonymized.

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Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used OpenAI Codex to assist with language editing, consistency checks, and the drafting of Python/Matplotlib code. The authors subsequently reviewed and edited all outputs, verified the technical statements and data mappings, and take full responsibility for the content of the manuscript.

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